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PulseTrack System Requirements Specification

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PulseTrack Health Management System

Software Requirements Specification (SRS)

Team Number: 1

Team Members:

- Christian Echavarria – Project Lead
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- Daniel Chan – Documentation Lead

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Version History

Version	Date	Author(s)	Change Comments
1.0	February 20, 2026	Christian Echavarria, Gahmliel Artison, Daniel Chan	Initial complete SRS derived from approved project proposal

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1. System Requirements

1.1 Functional Requirements

Table 2: Functional Requirements Summary

ID	Title	Priority
FR-1	User Authentication and Authorization	4
FR-2	Clinician Prescription Management	5
FR-3	Health Data Collection	5
FR-4	Local Data Storage	3
FR-5	Secure Synchronization	5
FR-6	Progress Monitoring and Visualization	2
FR-7	EHR Integration via HL7 FHIR	5

FR-1: User Authentication and Authorization

Title: Secure Access Control

Description:

The system shall provide secure authentication mechanisms for patients and clinicians using encrypted credentials and role-based authorization policies.

Priority: 5

Preconditions:

- User account is registered.
- The user has verified credentials.

Basic Flow:

1. User launches application.
2. The user enters credentials.
3. The system validates credentials against encrypted databases.
4. System assigns role-based permissions.
5. The user is directed to a role-specific dashboard.

Postconditions:

- Authenticated session established.
- Access rights enforced according to user role.

Use Case Diagram: Figure 2, Figure 3

FR-2: Clinician Prescription Management

Description:

Clinicians shall define, modify, and monitor measurable lifestyle prescriptions for patients.

Priority: 0

Preconditions:

- Clinician authenticated.
- Patient assigned to clinician.

Basic Flow:

1. The clinician selects the patient.
2. Defines activity or lifestyle metrics.
3. Sets thresholds and timeframes.
4. The system stores prescriptions.

Postconditions:

- Prescription available in patient dashboard.

FR-3: Health Data Collection

Description:

The system shall collect health metrics via smartphone sensors and manual user input.

Priority: 0

Preconditions:

- Patient authenticated.
- Sensor permissions granted.

Basic Flow:

1. System accesses accelerometer/GPS.
2. The patient optionally enters manual data.
3. Data validated and timestamped.

Postconditions:

- Health data stored securely.

FR-4: Local Data Storage

Description:

All patient data shall be stored locally by default using encrypted storage mechanisms.

Priority: 1

Postconditions:

- Data remains accessible offline.

FR-5: Secure Synchronization

Description:

The system shall synchronize data with secure cloud servers when enabled.

Priority: 1

Basic Flow:

1. Network connectivity detected.
2. Encrypted HTTPS channel established.
3. Data synchronized via REST/WebSockets.

FR-6: Progress Monitoring and Visualization

Description:

The system shall generate interactive dashboards displaying adherence trends and progress metrics.

Priority: 1

Postconditions:

- Graphical visualizations updated in real-time or upon sync.

FR-7: EHR Integration via HL7 FHIR

Description:

The system shall export standardized patient data using HL7 FHIR protocols.

Priority: 2

1.2 Non-Functional Requirements

Table 3: Non-Functional Requirements Summary

ID	Title	Priority
NFR-1	Security & Privacy	0
NFR-2	Performance	1
NFR-3	Availability	1
NFR-4	Usability	2
NFR-5	Scalability	2

NFR-1: Security & Privacy

All data shall be encrypted at rest and in transit in compliance with HIPAA guidelines.

Priority: 0

Applicable FR(s): FR-1 through FR-7

NFR-2: Performance

System response time shall not exceed 2 seconds for 95% of user interactions.

Priority: 1

NFR-3: Availability

The system shall support offline functionality with deferred synchronization.

Priority: 1

NFR-4: Usability

The system shall adhere to accessibility standards and user-centered design principles.

Priority: 2

NFR-5: Scalability

Server infrastructure shall support horizontal scaling using load balancing (NGINX).

Priority: 2

2. System Constraints

Table 4: System Constraints Overview

Category	Description	Priority
Tool Constraints	Android/iOS SDKs, Prisma DB, Node.js, REST, WebSockets, NGINX	1
Language Constraints	Swift, Android-compatible language, JavaScript, SQL	1
Platform Constraints	iOS and Android only	0
Hardware Constraints	Smartphone with accelerometer and GPS	1
Network Constraints	HTTPS required for sync	1
Deployment Constraints	App Store & Google Play compliance	1
Transition & Support	Documentation & training required	2
Budget & Schedule	$117,500 – $289,150 first year	0
Miscellaneous	HL7 FHIR compliance required	1

3. Requirements Modeling

Figure 1: High-Level System Architecture Diagram

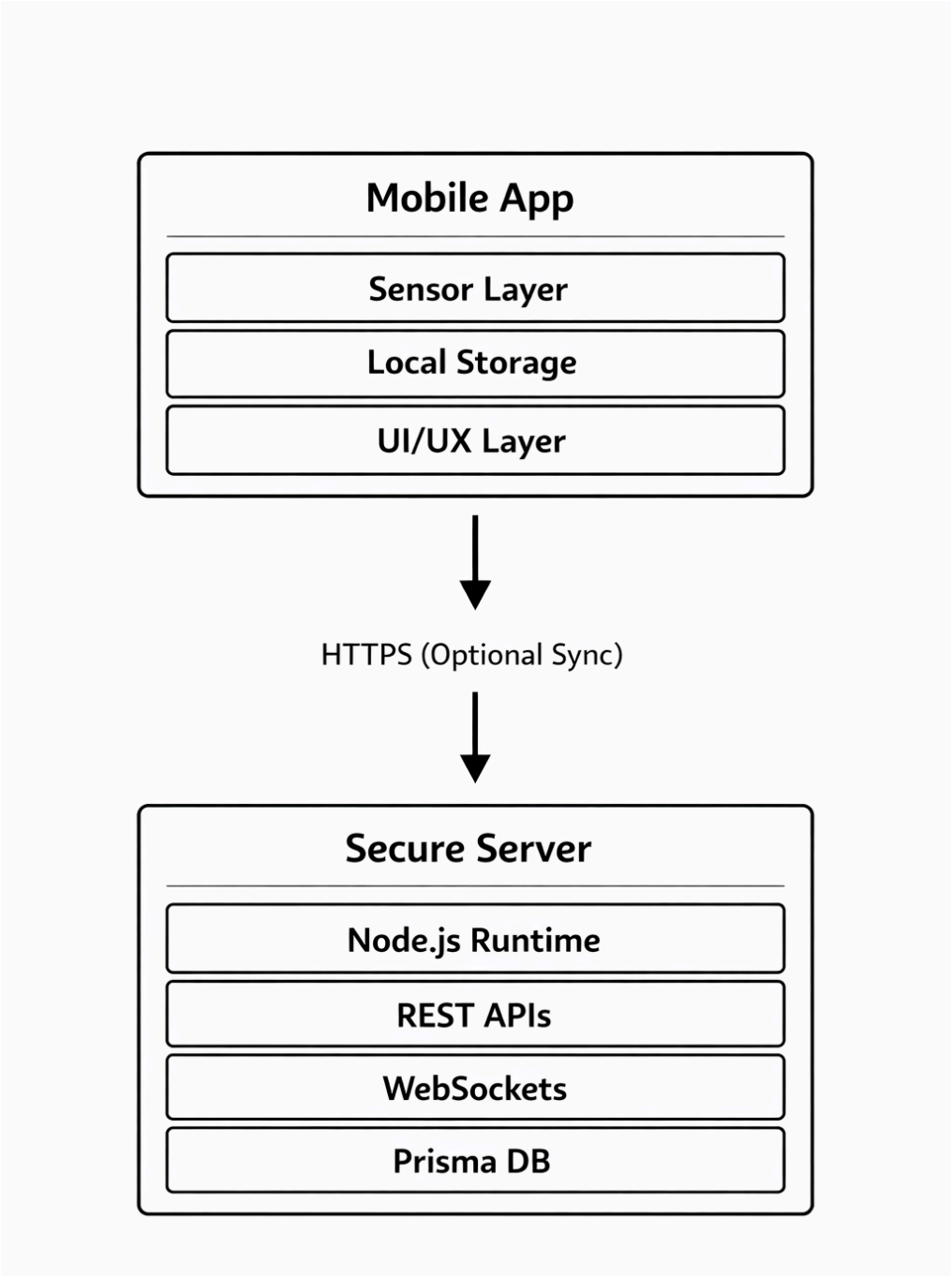


Figure 2: Patient Use Case Diagram

Actors: Patient

Use Cases:

- Login
- View Prescription
- Enter Health Data
- View Progress Dashboard
- Sync Data

Figure 3: Clinician Use Case Diagram

Actors: Clinician

Use Cases:

- Login
- Create Prescription
- Modify Prescription
- Review Patient Metrics
- Export Data

Figure 4: Data Flow Diagram (Level 1)

Patient Input → Validation → Encrypted Local Storage → (Optional) Secure Sync → Server DB → Clinician Dashboard

Table 7: Requirements Traceability Matrix

Requirement	Related NFR	Related Constraint
FR-1	NFR-1	Tool Constraint
FR-2	NFR-4	Platform Constraint
FR-3	NFR-2	Hardware Constraint
FR-5	NFR-3	Network Constraint
FR-7	NFR-1	Miscellaneous Constraint

4. Evolutionary Requirements

4.1 Functional Evolutionary Requirements

EF-1: Predictive Analytics

Future integration of AI-based risk prediction models.

Priority: 3

EF-2: Wearable Integration

Integration with commercial wearables via APIs.

Priority: 3

4.2 Non-Functional Evolutionary Requirements

ENF-1: Advanced Security Certification

SOC 2 and extended healthcare compliance certification.

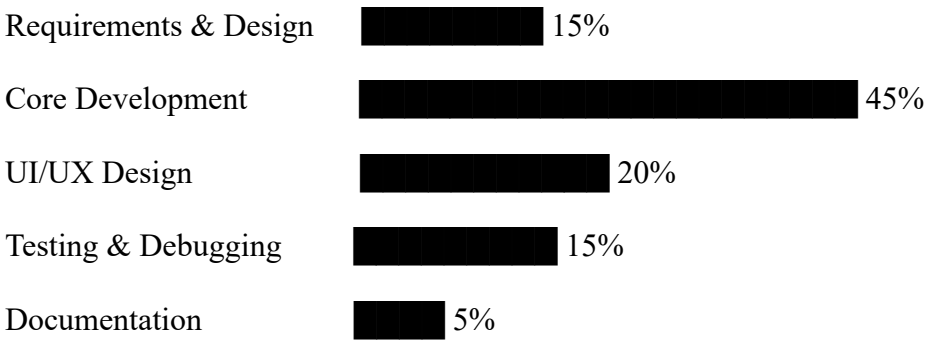
Priority: 3

Cost and Effort Distribution

Table 5: Cost Estimation Summary

Category	Estimated Cost (USD)
Effort Costs	$82,800 – $197,850
Hardware & Software	$34,700 – $91,300
Total First Year	$117,500 – $289,150

Figure 5: Projected Development Effort Distribution Graph



Risk Analysis

Table 6: Risk Analysis Matrix

Risk	Likelihood	Impact	Mitigation
Data Breach	Medium	High	Encryption & local-first storage
Low User Adherence	Medium	Medium	Feedback visualization
API Integration Failure	Low	Medium	Standardized APIs
Regulatory Delays	Low	High	HIPAA & FHIR compliance planning

Meeting Reports

Date	Participants	Topics Discussed	Decisions Made
Feb 5, 2026	Team	Initial Scope	Chronic disease focus
Feb 10, 2026	Team	Architecture	Local-first storage
Feb 15, 2026	Team	Security	Mandatory encryption

Conclusion

The PulseTrack Health Management System is designed as a secure, scalable, and interoperable mobile-first platform designed to bridge the gap between episodic clinical encounters and continuous patient-centered health monitoring. The system prioritizes data privacy, clinician-guided prescriptions, and adaptive synchronization strategies to support both offline and connected healthcare environments.

This SRS establishes a rigorous technical and operational foundation aligned with healthcare regulatory standards, modern mobile architecture practices, and long-term extensibility considerations.